INCH-POUND

MIL-DTL-82960B (OS)
11 August 2010
SUPERSEDING
MIL-DTL-82960A (OS)
16 April 2003

DETAIL SPECIFICATION

# INITIATOR, PROPELLANT ACTUATED, JAU-27/A (MF72)

*This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.*

## 1. SCOPE

**1.1 Scope.** This specification covers the requirements for the manufacture and acceptance of the JAU-27/A Propellant Actuated Initiator, referred to herein as "initiator".

## 2. APPLICABLE DOCUMENTS

**2.1 General.** The documents listed in this section are specified in sections 3, 4 or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents specified in Sections 3, 4 or 5 of this specification, whether or not they are listed.

**2.2 Government documents**.

**2.2.1 Specifications and standards.** The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SPECIFICATIONS

DEPARTMENT OF DEFENSE SPECIFICATIONS

| | |
|---|---|
| MIL-L-757 | Lead Styphnate |
| MIL-L-3055 | Lead Azide |

Comments, suggestions, or questions on this document should be addressed to Department of the Navy, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil/.

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| | |
|---|---|
| MIL-D-21625 | Design and Evaluation of Cartridges for Cartridge Actuated Devices |
| MIL-P-20444 | Primer, Percussion, M42 Parts for and Loading, Assembling and Packing |

STANDARDS

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-STD-464 | Electromagnetic Environmental Effects Requirements from Systems |
| MIL-STD-810 | Environmental Test Methods and Engineering Guidelines |
| MIL-STD-1168 | Ammunition Lot Numbering and Ammunition Data Card |

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia PA 19111- 5094.)

**2.2.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

SPECIFICATIONS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 53711)

| | |
|---|---|
| WS 5003 | HNS Explosive |

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

| | |
|---|---|
| DL 850AS131 | Initiator, Propellant Actuated, JAU-27/A |
| 850AS815 | Vibration Fixture, Initiator, Propellant Actuated, JAU-27/A |
| 850AS818 | Test Fixture: Initiator, Propellant Actuated, JAU-27/A |

(Application for copies should be addressed to the DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E11G7, Document Control, 4072 North Jackson Road Suite 106, Indian Head, MD 20640-5115 OFFICIAL BUSINESS.)

**2.3 Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are thosecited in the solicitation or contract.



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AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQC Z1.4 Sampling Procedures and tables for Inspection by Attributes

(Copies of this document are available online at http://webstore.ansi.org/ansidocstore/default.asp, or from the American Society for Quality, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

ASTM E 545 Standard Method for Determining Image Quality in Direct Thermal Neutron Radiographic Examination (DoD adopted)

ASTM E 1742 Standard Practice for Radiographic Inspection (DoD adopted)

(Copies of this document are available online at http://www.astm.org/ or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SOCIETY OF AUTOMOTIVE ENGINEER (SAE)

SAE AS 5604 Compass, Magnetic, Pilot's Standby (DoD adopted)

(Copies of this document are available online at http://www.sae.org/, or from the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, Pennsylvania, United States, 15096-001.)

**2.4 Order of precedence.** Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified (see 6.3), a sample shall be subjected to first article inspection in accordance with 4.2.1.

**3.2 Primary components.** Only one lot of primary components shall be used in each lot of initiators, although, a primary component lot may be used in more than one initiator lot.

**3.2.1 OEA P/N 2912100-1.** The primary components for the OEA P/N 2912100-1 initiator are: the M42C1-979 percussion primer per MIL-P-20444, lead styphnate per MIL-L-757, lead azide per MIL-L-3055, and hexanitrostilbene (HNS) Type II, grade “A” per WS 5003.

**3.2.1.1 M42C1-979 percussion primer.** The M42C1-979 Percussion Primer used in the OEA P/N 2912100-1 initiator shall be manufactured and tested in accordance with MIL-P-20444. The primers shall be loaded into the initiators within 24 months of new primer acceptance testing or within 12 months of primer recertification testing.



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**3.2.2 SCOT P/N 6034100.** The primary components for the SCOT P/N 6034100 initiator are: either the PVU-1/A or the M42C1-979 percussion primer per MIL-P-20444, lead azide per MIL-L-3055, and the hexanitrostilbene (HNS) Type II, grade "A" per WS5003.

**3.2.2.1 PVU-1/A or the M42C1-979 percussion primer.** The PVU-1/A or the M42C1-979 Percussion Primer used in the SCOT P/N 6034100 initiator, shall be manufactured and tested in accordance with MIL-P-20444. The primers shall be loaded into the initiators within 24 months of new primer acceptance testing or within 12 months of primer recertification testing.

**3.3 Initiator production.** The initiators shall be manufactured in accordance with DL 850AS131 and the documents listed thereon. Each explosive train subassembly and initiator shall meet the requirements of 3.3.1 through 3.3.1.3, and 3.3.2 through 3.3.2.5, respectively. Failure of any explosive train subassembly or initiator to meet the applicable requirements of 3.3.1 through 3.3.1.3 and 3.3.2 through 3.3.2.5 shall result in rejection of that unit. Failure to meet the requirements of 3.3.1.4 shall be cause for rejection of the entire lot of explosive train subassemblies.

**3.3.1 Explosive train assembly (OEA PN 2912103-1 or Scot PN 6028110-01) inspection and performance.**

**3.3.1.1 Visual inspection.** All explosive train assemblies manufactured under contract shall be visually inspected and screened for defects as outlined in 4.3.3.

**3.3.1.2 Leakage.** Explosive train assemblies shall not exhibit a leak rate in excess of $1 \times 10^{-5}$ cubic centimeters per second ($cm^3/s$) of air when tested in accordance with 4.3.4.

**3.3.1.3 Radiographic inspection.** Each explosive train assembly shall be examined radiographically in accordance with 4.3.5.1. Presence of the propellant, non-metal parts, no foreign hydrogenous materials and the proper assembly of pyrotechnic/non-metallic materials will be ensured. Any observable imperfections in assembly will be cause for rejection.

**3.3.1.4 Output gap.** Explosive train subassemblies shall detonate the test tip high order when tested as described in 4.3.6.

**3.3.2 Initiator inspection.**

**3.3.2.1 Firing pin.** Free movement, proper alignment and protrusion of the firing pin shall be verified during assembly of each firing mechanism produced under contract in accordance with 4.3.1.

**3.3.2.2 Firing mechanism.** Firing mechanisms from each lot, selected in accordance with 4.3.2, shall meet the requirements listed below when subjected to the test prescribed in 4.3.2.1 and 4.3.2.2.

**3.3.2.2.1 Non-actuation.** Firing mechanisms shall not actuate when a pressure of 400 −25, +0 psig is applied for 1 minute at the inlet port. See 4.3.2.1.

**3.3.2.2.2 Actuation.** Firing mechanisms not previously subject to the non-actuation pressure in accordance with 3.3.2.2.1 shall actuate when a pressure of 600 +25, -0 psig is applied (see 4.3.2.2).



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**3.3.2.3 Visual inspection.** All initiators manufactured under contract shall be visually inspected and screened for defects as outlined in 4.3.3.

**3.3.2.4 Radiographic inspection.** Initiators shall show proper assembly, presence of parts, and sealing when examined radiographically in accordance with 4.3.5.2.

**3.3.2.5 Demagnetization.** Initiators shall not deflect an approved Pilot's standby compass more than three degrees in either direction when passed in front of the compass. The initiators shall be tested in accordance with 4.3.7.

**3.3.2.6 Electrical bonding.** The bonding resistance of the initiator across the electrical bonded joint shall not be greater than 0.0025 ohm when tested in accordance with 4.3.8.

**3.3.2.7 Threaded joint torque.** The breakaway torque of the initiator shall be capable of withstanding a minimum breakaway torque of 200 in-lbs., without loosening, when tested in accordance with 4.3.9.

**3.4 Initiator performance.** Initiators shall be capable of meeting all requirements of 3.4.1 through 3.4.4.4.

**3.4.1 Vibration.** There shall be no evidence of deformation or derangement of components. The initiator shall not autoignite, shall be safe to handle, and shall meet the requirements of 3.3.2.4 and 3.4.4 following vibration testing of 4.3.10.

**3.4.2 Temperature and humidity (T & H) cycling.** The initiators shall meet the requirements of 3.3.2.4 and 3.4.4 after being exposed to the T & H cycle in accordance with 4.3.11.

**3.4.3 Shock.** The initiators shall meet the requirements of 3.3.2.4 and 3.4.4 following shock testing of 4.3.9.

**3.4.4 Ballistic requirements.** Initiator(s) shall meet the requirements of 3.4.4.1 through 3.4.4.4 when tested in accordance with 4.3.13.

**3.4.4.1 Shear pin actuation pressure.** The upper (U) and lower (L) limits for shear pin actuation pressure (psig) when fired over the temperature range from –65°F to +200°F shall be U= 600 psig and L = 400 psig in accordance with 4.3.10.1.

**3.4.4.2 Actuation time.** The limit for actuation time (milliseconds) shall not be greater than 10 milliseconds when evaluated in accordance with 4.3.10.2.

**3.4.4.3 High order detonation.** The minimum plate dent for high order detonation of the output SMDC test tip on the aluminum witness plate shall be 0.001 inch when evaluated in accordance with 4.3.10.3.

**3.4.4.4 Misfire.** There shall be no misfires (see 4.3.10.4).



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## 4. VERIFICATION

**4.1 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Production inspection (see 4.2.2)
c. Lot acceptance inspection (see 4.2.3)

**4.2 Types of inspections.**

**4.2.1 First article inspection**. Before entering into quantity production, an acceptable first article sample of 54 initiators shall be prepared using the same procedures and methods proposed for the production lot. A first article sample acceptable for functional and environmental testing shall be defined as a sample which has met the applicable requirements of 3.2 through 3.3 of this specification. All fifty-four initiators shall be expended in the tests (see table I, below). The contractor shall deliver the initiators to the activity designated by the contracting agency for tests. Any damage inflicted by the environmental treatments, which would adversely affect the performance of the item in service application shall cause rejection of the first article sample. Any further production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor's risk. Failure of any initiator to comply with the requirements of section 3 shall be cause for rejection of the first article represented.

TABLE I. Initiator first article lot test plan

| Test Sequence | Requirement Paragraph | Test Paragraph | Sample Group | | | |
|---|---|---|---|---|---|---|
| | | | I | II | III | IV |
| Visual | 3.3.2.3 | 4.3.3 | 6 | 18 | 6 | 24 |
| Radiographic | 3.3.2.4 | 4.3.5 | 6 | 18 | 6 | 24 |
| Electrical Bonding | 3.3.2.6 | 4.3.8 | 6 | 18 | 6 | 24 |
| Threaded Joint Torque | 3.3.2.7 | 4.3.9 | 6 | 18 | 6 | 24 |
| Vibration | 3.4.1 | 4.3.10 | 6 | | | |
| T & H Cycling | 3.4.2 | 4.3.11 | | 12 | | |
| Shock | 3.4.3 | 4.3.12 | | | 6 | |
| Radiographic | 3.3.2.4 | 4.3.5.2 | 6 | 12 | 6 | |
| Ballistic test | 3.4.4 | 4.3.13 | | | | |
| −65 ± 5°F | | | 2 | 6 | 2 | 8 |
| 70 ± 5°F | | | 2 | 6 | 2 | 8 |
| 200 ± 5°F | | | 2 | 6 | 2 | 8 |

**4.2.2 Production inspection.** All production initiators manufactured under contract shall be inspected and screened to requirements specified in 3.2 through 3.3 and table II. Any unit failing to meet the



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requirements listed in 3.2 through 3.3.1.3 and 3.3.2.1 through 3.3.2.5 shall be rejected and removed from the lot of explosive train subassemblies or initiators, where applicable.

**4.2.3 Lot acceptance inspection.** Lot acceptance inspection shall consist of the examinations and tests specified in table III. Failure of any sample initiator to comply with the requirements listed in table III shall be cause for rejection of the lot represented.

**4.2.3.1 Sample size.** A random sample from each explosive train subassembly lot shall be selected in accordance with table IV for the output gap test. A random sample from each initiator production lot, including the initiator retained sample for investigative purposes, shall be selected in accordance with table V for lot acceptance inspection. Test samples and initiators retained for investigative purposes shall not be applied as part of the quantity specified for delivery by the contract or purchase order.

TABLE II. Explosive train subassembly inspections and tests.

| Inspection/Test | Requirement Paragraph | Test Paragraph | Quantity |
|---|---|---|---|
| Visual | 3.3.1.1 | 4.3.3 | All |
| Leakage | 3.3.1.2 | 4.3.4 | All |
| Radiographic | 3.3.1.3 | 4.3.5.1 | All |
| Output Gap test<br>75 ±15°F | 3.3.1.4 | 4.3.6 | Test Sample |

TABLE III. Initiator lot acceptance inspections and tests.

| Inspection/Test | Requirement Paragraph | Test Paragraph | Quantity |
|---|---|---|---|
| Visual | 3.3.2.3 | 4.3.3 | All |
| Radiographic | 3.2.2.4 | 4.3.5 | All |
| Electrical Bonding | 3.3.2.6 | 4.3.8 | Test and retain sample |
| Threaded Joint Torque | 3.3.2.7 | 4.3.9 | Test and retain sample |
| Ballistic Test<br>65 ± 5°F<br>70 ± 5°F<br>200 ± 5°F | 3.4.4 | 4.3.13 | <br>1/3 test sample<br>1/3 test sample<br>1/3 test sample |



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TABLE IV. Explosive train subassembly lot acceptance sampling.

| Lot Size | Retained Sample Size |
|---|---|
| 2 to 50 | 2 |
| 51 to 100 | 4 |
| 101 to 150 | 6 |
| 151 to 280 | 8 |
| 281 to 500 | 10 |
| 501 to 1200 | 12 |

TABLE V. Initiator lot acceptance sampling.

| Lot Size | Test Sample Size | Retained Sample Size |
|---|---|---|
| 2 to 50 | 9 | 3 |
| 51 to 100 | 12 | 3 |
| 101 to 150 | 21 | 3 |
| 151 to 280 | 33 | 3 |
| 281 to 500 | 51 | 3 |
| 501 to 1200 | 80 | 3 |

**4.3 Inspections and tests.**

**4.3.1 Firing pin assembly inspection.** The firing pin used in each initiator manufactured under contract shall be subjected the tests described in 4.3.1.1 and 4.3.1.2 prior to and during assembly of the firing mechanism. Parts not satisfying these requirements shall be discarded.

**4.3.1.1 Firing pin movement.** The firing pin, without O-ring, shall move freely under its own weight throughout maximum travel provided for the pin in the firing mechanism.

**4.3.1.2 Alignment.** Alignment and protrusion of the firing pin striker shall be verified.

**4.3.2 Firing mechanism test.** From each lot, a number of firing mechanisms shall be selected in accordance with Table I, General Inspection Level I, of ASQC Z1.4 with the additional requirement that the sample size used for non-actuation and actuation testing shall not be less than eight units. If the sample is not equally divisible, the excess initiator shall be placed in the group to be tested at –65°F. The firing mechanism will be mated to an inert explosive train cartridge to perform testing.



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**4.3.2.1 Non-actuation.** Half of the samples chosen for this test shall be conditioned for at least 4 hours but not more than 24 hours at –65°±5°F. The remaining half shall be conditioned for at least 4 hours but not more than 24 hours at 200°F ± 5°F. Within three minutes after removal from the conditioning chamber a pressure of 400 +0, –25 psig shall be applied to the inlet port of each firing mechanism for 1 minute. Actuation of any firing mechanism shall result in rejection of the lot.

**4.3.2.2 Actuation.** Samples will be chosen as outlined in paragraph 4.3.2. Half the samples shall be conditioned to –65°F ± 5°F for at least 4 hours but not more than 24 hours. The other half of the samples shall be conditioned to +200°F ±5°F for at least 4 hours but not more than 24 hours. Within three minutes after removal from the conditioning chamber, a pressure of 600 +25, –0 psig shall be applied to the inlet port of each firing mechanism. Failure of any firing mechanism to actuate shall result in rejection of the lot. Proper actuation is defined as the shearing of the shear pin, followed by free travel and bottoming of the firing pin against the head of an inert cartridge.

**4.3.2.3. Rework.** If the cause of failure in 4.3.2.1 and 4.3.2.2 can be corrected by rework of the lot, the lot in question may be reworked to correct the deficiency and a new sample, selected in accordance with Table I, General Inspection Level I, of ASQC Z1.4, submitted for test. Firing mechanisms which pass the tests of 4.3.2.1 and 4.3.2.2 may be reworked by replacement of the shear pins, O-rings, and damaged firing pins.

**4.3.3 Initiator and explosive train subassembly visual inspection.** Visually examine all explosive train subassemblies prior to initiator assembly. Each assembled initiator shall be visually examined. Reject those with inadequate or illegible descriptive markings or serial numbers, dents, deep scratches, or other defects which may prevent proper installation or function.

**4.3.4 Explosive train subassembly leak test.** Each explosive train subassembly produced under contract shall be leak tested in a dry gas leak tester to determine conformance with 3.3.1.2. Units which exhibit a leak rate in excess of $1 \times 10^{-5}$ standard $cm^3$/sec air at a pressure differential of 1.0 ± 0.1 atmosphere shall be rejected.

**4.3.5 Radiographic examination.**

**4.3.5.1 Explosive train subassemblies examination.** Radiographic examination of explosive train subassemblies shall include X-ray in accordance with ASTM E 1742 and N-ray in accordance with ASTM E 545. All explosive train subassemblies shall be identified with serial numbers beginning with 001 prior to examination. The initiators shall be arranged on trays or boards in consecutive numerical order with any missing serial numbers identified on the radiographic plate. The explosive train subassemblies shall be positioned for the most revealing exposure with the long axis perpendicular to the radiographic beam. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4 × 6 inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, the DODIC, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests (see 6.2). Defective assemblies found during radiographic review are to be marked on the radiographic plate and removed from the production lot.

**4.3.5.2 Initiator examination.** Radiographic examination of initiators shall include x-ray in accordance with ASTM E 1742. All initiators shall be identified with serial numbers beginning with 001 prior to examination. The initiators shall be arranged on trays or boards in consecutive numerical order with any



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missing serial numbers identified on the radiographic plate. The initiators shall be positioned for the most revealing exposure with the long axis perpendicular to the radiographic beam. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4 × 6 inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, the DODIC, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests (see 6.2). Defective initiator assemblies found during radiographic review are to be marked on the radiographic plate and removed from the production lot.

**4.3.6 Output gap test.** Prior to assembly of initiators, a random sample of explosive train subassemblies, selected per table IV, shall be subjected to output gap testing. Explosive train subassembly explosive output shall be capable of detonating SMDC test tips across an air gap which is four times the maximum production gap of the initiator.

Failure of any explosive train subassembly to meet the requirement of 3.3.1.4 shall result in rejection of the lot.

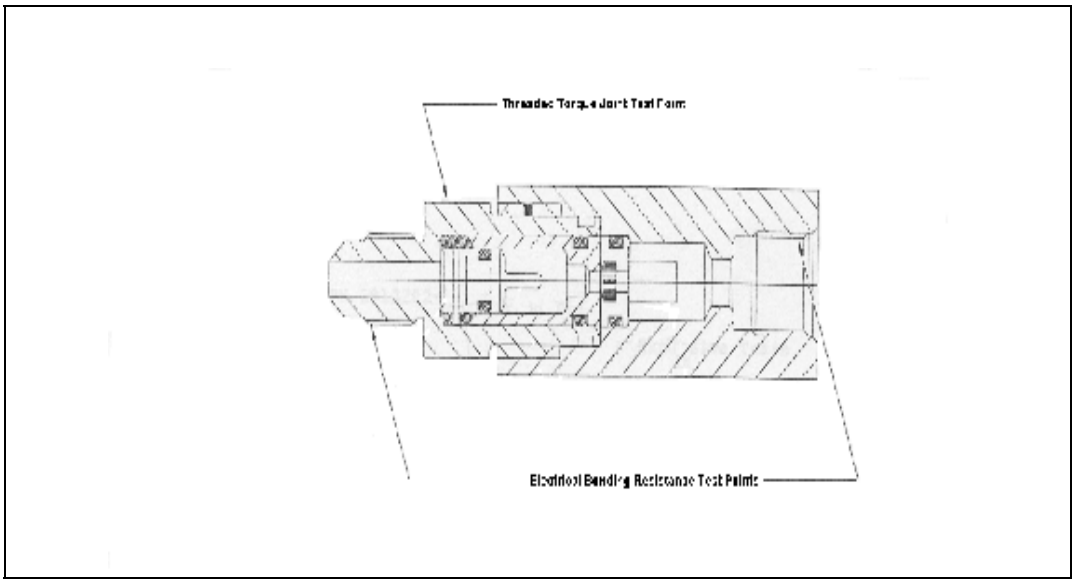
**4.3.7 Demagnetization.** Each initiator shall be moved past a pilot's standby compass face (in accordance with SAE AS 5604) by placing the initiator 6 inches from the compass set in an east-west heading. The initiator and the compass shall be placed at the same height from the floor. The initiators shall be moved past the compass face in a longitudinal direction for its entire length, rotated 90 degrees in the horizontal position about it's longitudinal axis, and moved back past the compass face. This procedure shall be repeated for each 90 degrees rotation of the initiator. The initiator shall be acceptable if the requirements of 3.3.2.5 are met.

**4.3.8 Electrical Bonding.** Sample initiators shall be electrically tested for resistance between the two end ports of the initiator, as shown in figure 1, in accordance with MIL-STD-464. The electrical bonding points shall be free from grease, paint, lacquer or other high resistance properties. Failure of any initiator to comply with the requirements of 3.3.2.6 shall be cause for rejection of the lot and/or first article.

**4.3.9 Threaded joint torque.** Sample initiators shall be tested for breakaway torque of threaded joints, see figure 1, using a calibrated wrench. Failure of any initiator to comply with the requirements of 3.3.2.7 shall be cause for rejection of the lot and/or first article.



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FIGURE 1. Threaded torque joint and electrical bonding resistance test points.

**4.3.10 Vibration test.**

**4.3.10.1 Non-gunfire vibration.** The test shall be conducted in accordance with MIL-STD-810B, Table 514.1-I Curve "AT" except that for each cycling period the time shall be divided equally for vibration at –65, 70, and 200°F. The initiators shall be mounted in the vibration test fixture, Drawing 850AS815, for the JAU-27/A initiator. The non-gunfire vibration test shall be performed on all six initiators. The six initiators shall be mounted on the test fixture at the same time and one shall be monitored for response during the test.

**4.3.10.2 Gunfire vibration.** The gunfire vibration test shall be performed on all six initiators, on each of the three orthogonal axes, and shall be conducted at 70°F test temperature only. The six initiators shall be mounted on the test fixture at the same time and one shall be monitored for response during the test.

**4.3.10.2.1 Resonance search.** A resonance search shall be conducted on each of the six initiators at 70°F in each of the three orthogonal axes. The frequency sweep shall be made from 50 Hz to 500 Hz and back to 50 Hz in approximately 7.50 minutes to the vibration test levels shown in table VI. The resonance points shall be noted. A resonance point is defined as a frequency which has a transmissibility (output/input) of two or greater. If more than three resonant points are noted from the resonance survey, only the three more severe will be used for resonance dwell vibration.



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TABLE VI. Vibration test levels.

| Frequency (Hz) | Acceleration or Amplitude |
|---|---|
| 50 to 54 | 0.36 inch D.A. |
| 55 to 210 | ± 5 G's |
| 211 to 300 | 0.00222 inch D.A. |
| 301 to 500 | ± 10 G's |

**4.3.10.2.2 Resonance dwell.** The six initiators shall be vibrated in each of three orthogonal axes at 70°F for 7.50 minutes at each resonant point (to a maximum of three) obtained during the resonance survey in 4.3.10.2.1 herein. The gunfire vibration test levels are shown in table VI depending upon the number of resonance frequencies determined in the resonance survey per 4.3.10.2. The remainder of the sweep time in accordance with table VII shall be used for vibration cycling per 4.3.10.2.3 herein.

**4.3.10.2.3 Vibration cycling.** All six initiators shall be subjected to sinusoidal vibration cycling at 70°F for a period of time as specified in 4.3.10.2.2 on each of the three orthogonal axes with the frequency varying logarithmically from 50 Hz to 500 Hz and back to 50 Hz in approximately 7.50 minutes. The gunfire vibration test levels shall be shown in table VI.

**4.3.10.2.4 Fixed dwell.** After sinusoidal vibration cycling, the six initiators shall be subjected to a series of fixed dwells as shown in table VII. The fixed narrow band dwells shall be performed by sweeping the frequency of ± 5 percent of the specified center frequency (e.g., the 100 Hz dwell should be performed from 95 to 105 Hz at the logarithmic sweep rate specified in 4.3.10.2.3).

**4.3.10.2.5 Sinusoidal sweep.** After completion of the fixed dwell testing, the six initiators shall be subjected to seven sinusoidal vibration sweeps in each of the three orthogonal axes. The frequency shall vary logarithmically from 500 Hz to 2000 Hz and back to 500 Hz. The sweep cycling rate shall be as specified in 4.3.10.2.3 herein, with an applied input of ± 10 G's.

**4.3.11 Temperature and humidity cycling.** The temperature and humidity cycling test shall be conducted in accordance with MIL-D-21625. Failure of any cartridge to comply with the requirements of 3.4.2 shall be cause for rejection of the lot or first article.

**4.3.12 Shock test.** The shock test shall be conducted in accordance with MIL-D-21625. Failure of any initiator to comply with the requirements of 3.4.3 shall be cause for rejection of the lot or first article.

**4.3.13 Ballistic test.** Sample initiators shall be conditioned for at least 4 hours but not more than 24 hours at the specified temperatures. The initiators shall be removed from the conditioning chamber and test fired within 3 minutes. If any initiator is not fired within 3 minutes after removal, it shall be reconditioned at the specified temperature for an additional 4 hours and then tested. The initiators shall be fired in the test fixture as shown in figure 2 (see Drawing 850AS818). Instrumentation shall be provided to measure and record the input air pressure (800 +25, –0 psig); shear pin actuation pressure (psig); actuation time



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(milliseconds); and the high order detonation of the output shielded mild detonating cord (SMDC) test tip on the aluminum witness plate. The initiators shall meet the requirements specified in 3.4.4.

TABLE VII. Gunfire sinusoidal vibration test schedule.

| Fixed Dwell (fd) at the following center frequencies | Number of Resonances | | | |
|---|---|---|---|---|
| | 0 | 1 | 2 | 3 |
| | Sweep Time | | | |
| | 30 | 22.5 | 15.0 | 7.50 |
| | Resonance Dwell Time | | | |
| | — | 7.5 | 15.0 | 22.50 |
| 67 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 100 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 135 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 200 Hz | 15.00 | 15.00 | 15.00 | 15.00 |
| 267 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 300 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 335 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 400 Hz | 15.00 | 15.00 | 15.00 | 15.00 |
| 467 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| 500 Hz | 7.50 | 7.50 | 7.50 | 7.50 |
| Total Time Each Axis | 120.0 | 120.0 | 120.0 | 120.0 |

Note: When an equipment resonance occurs within 5% of a fixed dwell frequency, the fixed dwell period shall be omitted and only the resonance dwell performed. The omitted fixed dwell time shall then be added to the sweep time.



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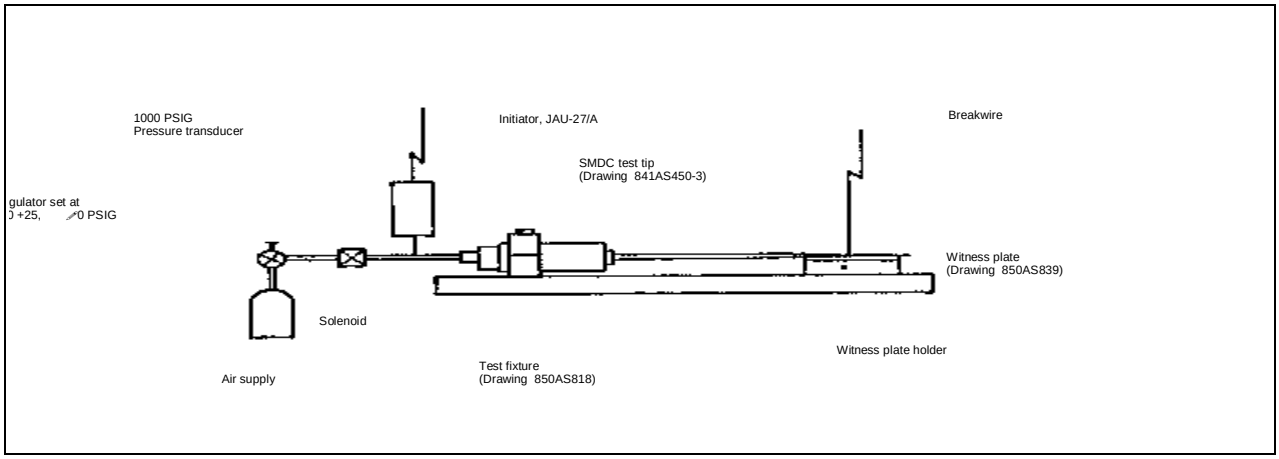



FIGURE 2. Test setup.

**4.3.13.1 Shear pin actuation pressure.** The shear pin actuation pressure is the gas pressure required to actuate the gas operated firing mechanism when a gas pressure of 800 +25, –0 psig is applied at a rate between 10,000 psig/sec and 50,000 psig/sec. The lot shall be acceptable with respect to shear pin actuation pressure if the ballistic sample meets the requirements of 3.4.4.1.

**4.3.13.2 Actuation time.** The actuation time shall be measured from the initiation of the inlet air pressure supply to the detonation of the outlet SMDC test tip. Each initiator shall meet the requirements of 3.4.4.2.

**4.3.13.3 High order detonation.** High order detonation of the output SMDC test tip shall be demonstrated in a 6061-T6 Aluminum witness plate. Each initiator shall meet the requirements of 3.4.4.3.

**4.3.13.4 Misfire.** An initiator that does not fire under the conditions specified in 4.3.10 shall be considered defective, and the lot shall be considered unacceptable, unless the misfire is clearly attributable to faulty test equipment or procedure. Each initiator shall meet the requirements of 3.4.4.4.

**4.3.13.5 Test failure.** If test failure is attributable to an assignable cause, excluding the test initiator, the original test results shall be discarded and that test reconducted.

## 5. PACKAGING

**5.1 Packaging.** For acquisition purposes, the packaging requirements will be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

**6.1 Intended use.** The initiator, propellant actuated, converts ballistic gas energy into the explosive energy required to initiate the shielded mild detonating cord (SMDC) used in the FA-18 A/B/C/D/E/F, F-14, TAV-8B, and T-45 aircraft. This propellant actuated initiator is military unique and there is no equivalent commercial application for this design.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1, 2.2.2, and 2.3).

c. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2.1).

d. Directions for shipping radiographs of entire production lot and marriage log (report indicating cartridge serial number to initiator serial number), along with the ballistic sample to the activity conducting the production lot acceptance tests (see 4.3.5.1 and 4.3.5.2).

e. That safety requirements are applicable (see 6.4).

**6.3 First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested (see 4.2.1). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

**6.3.1 Bids.** Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

**6.4 Ammunition and explosive safety.**

**6.4.1 Contractor's responsibility.** The safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable when this specification is used for acquisition.



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**6.4.2 Government activity's responsibility.** The safety precaution requirements of "Ammunition and Explosives Ashore", OP5 are applicable when this specification is used as part of a description of work to be accomplished by a government activity.

**6.5 Subject term (key word) listing.**

Aircraft, military
Shielded mild detonating cord
SMDC

**6.6 Changes from previous issue.** Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:
Navy-OS

Preparing activity
Navy - OS
(Project 1377-2010-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.



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